Chapter 7 — Swaps

$$\text{Gain by swap} = \Delta_{\text{fixed}} - \Delta_{\text{floating}}$$

Chapter 8 — Option Payoffs

$$\begin{aligned}
\text{Long CE} &= \max(S_T - K,\ 0)\\
\text{Short CE} &= -\max(S_T - K,\ 0) = \min(K - S_T,\ 0)\\
\text{Long PE} &= \max(K - S_T,\ 0)\\
\text{Short PE} &= -\max(K - S_T,\ 0) = \min(S_T - K,\ 0)
\end{aligned}$$

Chapter 9 — Bounds & Put-Call Parity

Bounds

$$\begin{aligned}
&\max(S_0 - Ke^{-rT}, 0) \le c \le C \le S_0\\
&\max(Ke^{-rT} - S_0, 0) \le p \le Ke^{-rT}\\
&\max(K - S_0, 0) \le P \le K\\
&c \ge \max(S_0 - D - Ke^{-rT}, 0)\\
&p \ge \max(D + Ke^{-rT} - S_0, 0)
\end{aligned}$$

Put-Call Parity — European

$$\begin{aligned}
&c + Ke^{-rT} = p + S_0\\
&c + D + Ke^{-rT} = p + S_0
\end{aligned}$$

Put-Call Parity — American

$$\begin{aligned}
&S_0 - K \le C - P \le S_0 - Ke^{-rT}\\
&S_0 - D - K \le C - P \le S_0 - Ke^{-rT}
\end{aligned}$$

Chapter 10 — Spreads & Combinations (Payoffs at Expiry)

Bull Spread (long call K_1, short call K_2, $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} 0 & S_T < K_1\\ S_T - K_1 & K_1 \le S_T < K_2\\ K_2 - K_1 & S_T \ge K_2 \end{cases}$$

Bear Spread (long put K_2, short put K_1, $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} K_2 - K_1 & S_T \le K_1\\ K_2 - S_T & K_1 < S_T < K_2\\ 0 & S_T \ge K_2 \end{cases}$$

Box Spread (bull call spread + bear put spread, $K_1 < K_2$)

$$\text{Payoff} = K_2 - K_1 \qquad \forall\ S_T$$

Butterfly Spread (long K_1, short $2 \times K_2$, long K_3; $K_1 < K_2 < K_3$)

$$\text{Payoff} = \begin{cases} 0 & S_T \le K_1\\ S_T - K_1 & K_1 < S_T \le K_2\\ K_3 - S_T & K_2 < S_T < K_3\\ 0 & S_T \ge K_3 \end{cases}$$

Max payoff $= K_2 - K_1$ at $S_T = K_2$ (symmetric wings)

Straddle (long call + long put, same strike K)

$$\text{Payoff} = \begin{cases} K - S_T & S_T < K\\ 0 & S_T = K\\ S_T - K & S_T > K \end{cases}$$

Strangle (long put K_1, long call K_2, $K_1 < K_2$)

$$\text{Payoff} = \begin{cases} K_1 - S_T & S_T < K_1\\ 0 & K_1 \le S_T \le K_2\\ S_T - K_2 & S_T > K_2 \end{cases}$$

Chapter 13 — Black-Scholes-Merton

BSM Differential Equation

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

Pricing Formulas

$$\begin{aligned}
c &= S_0 N(d_1) - Ke^{-rT}N(d_2)\\
p &= Ke^{-rT}N(-d_2) - S_0 N(-d_1)\\
d_1 &= \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}\\
d_2 &= \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}
\end{aligned}$$

Chapter 17 — The Greek Letters

$$\begin{aligned}
\Delta_{\text{call}} &= N(d_1) & \Delta_{\text{put}} &= N(d_1) - 1\\
\Gamma &= \frac{N'(d_1)}{S_0\sigma\sqrt{T}} & \nu &= S_0\sqrt{T}\,N'(d_1)\\
\rho_{\text{call}} &= KTe^{-rT}N(d_2) & \rho_{\text{put}} &= -KTe^{-rT}N(-d_2)
\end{aligned}$$

$$\begin{aligned}
\Theta_{\text{call}} &= -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)\\
\Theta_{\text{put}} &= -\frac{S_0 N'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2)
\end{aligned}$$

$$\begin{aligned}
&N'(x) = \frac{1}{\sqrt{2\pi}}\,e^{-x^2/2}\\
&\Delta\Pi \approx \Theta\,\Delta t + \tfrac{1}{2}\Gamma\,(\Delta S)^2\\
&\Theta + rS\Delta + \tfrac{1}{2}\sigma^2 S^2\Gamma = r\Pi
\end{aligned}$$

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997